# CMPG 311 2026- Group 3

## Real Estate Property Listing System

## Phase 2

***Database Design***

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| **Module** | CMPG 311 – Database Systems |
| **Phase** | Phase 2 – Database Design |
| **Topic** | Real Estate Property Listing System |
| **Submission Date** | 24 April 2026 |
| **Institution** | North-West University (NWU) |
| **Year** | 2026 |

# Conceptual Design

## 1.Business Rules:

1.A branch needs to have at least one agent on it; each agent should only belong to one branch.

2. An agent can take care of zero or many property listings. Each listing can only be handled by one agent.

3.Each property has one and only one property type. A property type can classify zero or many properties.

4. Each property must be in one and only one location. A location can include zero or many properties.

5. A location has only one postal code, city, and province, but it can encompass zero or many properties.

6. A client may want to save zero or many properties, but each property is only saved by zero or many clients. This is an M:N relationship implemented with the CLIENT_SAVED_PROPERTY bridge entity.

7.Each row in CLIENT_SAVED_PROPERTY has only one ClientID and only one property ID to identify the row. The composite primary key is a combination of the ClientID and propertyID.

8. A client may send zero or many inquiries; each inquiry can only be sent by one client.

9. An inquiry has only one property and a property may have zero or many inquiries, but still only one is required.

10.Each inquiry must be attended to by one agent; an agent can handle zero or many inquiries.

11.Each property will be associated with one and only one transaction; a transaction can only be involved with zero or many properties but one is required.

12. A client can be a participant of zero or many transactions; each transaction involves one and only one client.

13. An agent may close zero or many transactions; each transaction is only closed by one agent.

14.Each property may have zero or many features, but only one feature can be associated with each feature record.

# 2. Entity Relationships

## Relationships Summary

| Entity A | Relationship | Entity B | Cardinality | Participation |
|---|---|---|---|---|
| BRANCH | employs | AGENT | **1 : M** | Branch: **mandatory** Agent: **mandatory** |
| AGENT | manages | PROPERTY | **1 : M** | Agent: **optional** Property: **mandatory** |
| LOCATION | | PROPERTY | **1: M** | Location: **Mandatory** Property: **Optional** |
| PROPERTY_TYPE | classifies | PROPERTY | **1 : M** | Property Type: **optional** Property: **mandatory** |
| CLIENT | submits | INQUIRY | **1 : M** | Client: **optional** Inquiry: **mandatory** |
| PROPERTY | receives | INQUIRY | **1 : M** | Property: **optional** Inquiry: **mandatory** |
| AGENT | handles | INQUIRY | **1 : M** | Agent: **optional** Inquiry: **mandatory** |
| PROPERTY | involved in | TRANSACTION | **1 : 1** | Property: **optional** Transaction: **mandatory** |
| CLIENT | party to | TRANSACTION | **1 : M** | Client: **optional** Transaction: **mandatory** |
| PROPERTY | Has | PROPERTY_FEATURE | **1:M** | Property: Mandatory |

| | | | | |
|---|---|---|---|---|
| | | | | Feature: **Optional** |
| AGENT | closes | TRANSACTION | **1 : M** | Agent: **optional** Transaction: **mandatory** |
| CLIENT | saves | PROPERTY | **M : M → bridge** | **Both optional** |

## 3.Advanced Design Features

The ERD includes the following advanced features:

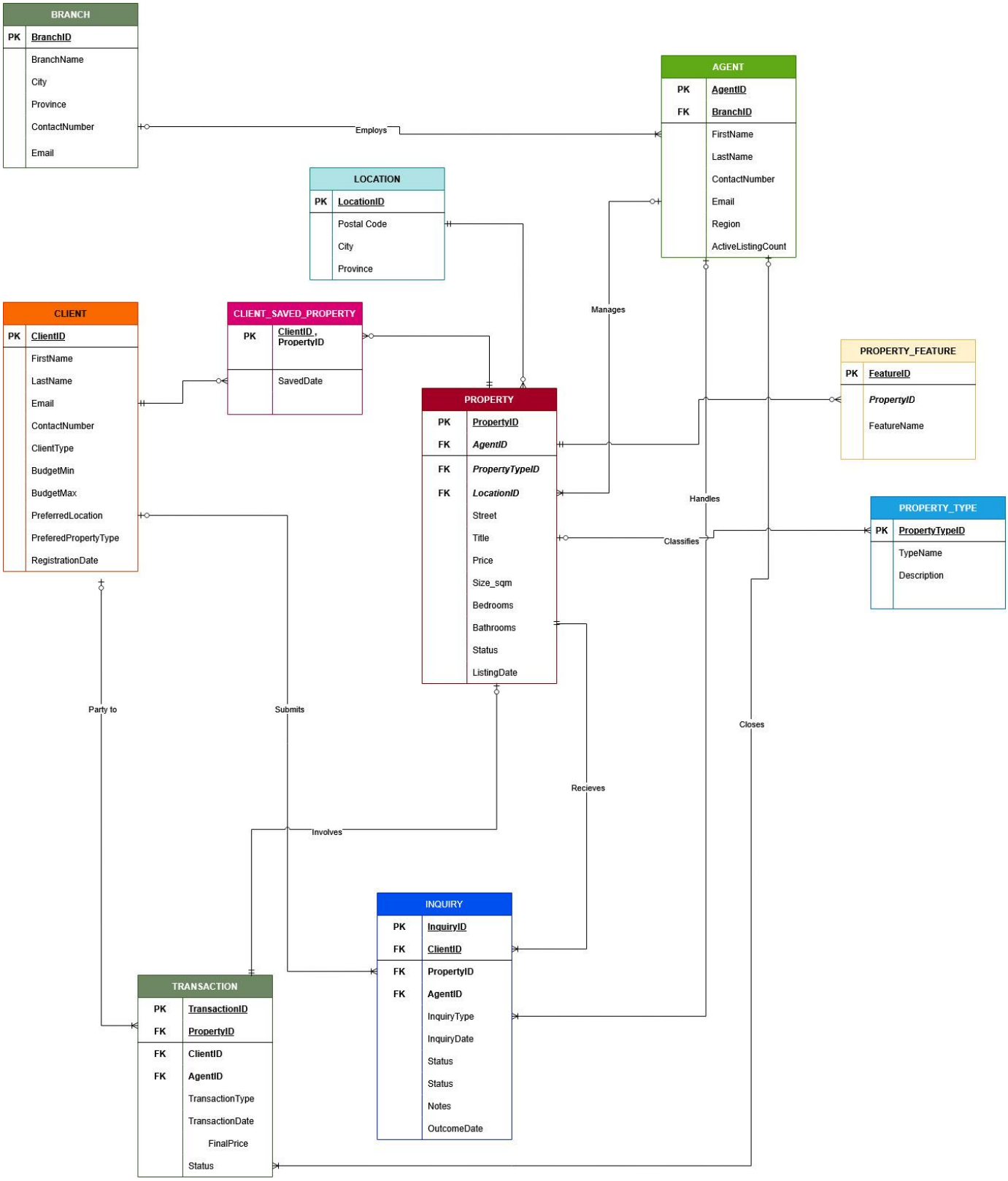
1. **Composite attribute**
   - Property address broken into Street, City, Province — stored as separate attributes but logically one composite address.
2. **Multivalued attribute**
   - Features (pool, garage, garden, security) — a property can have many features. Stored in a separate PROPERTY_FEATURE table.

3. **Derived attribute**
   - ActiveListingCount on AGENT — calculated from the count of PROPERTY records linked to that agent with status = 'available'.
4. **Composite key (bridge entity)**
   - CLIENT_SAVED_PROPERTY uses (ClientID + PropertyID) as a composite primary key — neither alone is unique.
5. **Mandatory / optional relationships**
   - Every property MUST have an agent (mandatory); an agent MAY have zero properties (optional). Show with single/double lines on your diagram.
6. **Weak entity**

- INQUIRY could be modelled as a weak entity — it has no meaning without a CLIENT and PROPERTY, and relies on them for identification.

7. Strong / weak relationships
   - AGENT–BRANCH is a strong relationship (both exist independently). INQUIRY–CLIENT is a weak relationship (inquiry depends on client)

# 4.The Entity Relationship Diagram (ERD)


BRANCH
PK BranchID
BranchName
City
Province
ContactNumber
Email
Employs
AGENT
PK AgentID
FK BranchID
FirstName
LastName
ContactNumber
Email
Region
ActiveListingCount
LOCATION
PK LocationID
Postal Code
City
Province
Manages
CLIENT
PK ClientID
FirstName
LastName
Email
ContactNumber
ClientType
BudgetMin
BudgetMax
PreferredLocation
PreferedPropertyType
RegistrationDate
CLIENT_SAVED_PROPERTY
PK ClientID , PropertyID
SavedDate
PROPERTY_FEATURE
PK FeatureID
PropertyID
FeatureName
PROPERTY
PK PropertyID
FK AgentID
FK PropertyTypeID
FK LocationID
Street
Title
Price
Size_sqm
Bedrooms
Bathrooms
Status
ListingDate
Handles
Classifies
PROPERTY_TYPE
PK PropertyTypeID
TypeName
Description
Party to
Submits
Closes
Recieves
Involves
INQUIRY
PK InquiryID
FK ClientID
FK PropertyID
FK AgentID
InquiryType
InquiryDate
Status
Status
Notes
OutcomeDate
TRANSACTION
PK TransactionID
FK PropertyID
FK ClientID
FK AgentID
TransactionType
TransactionDate
FinalPrice
Status

# Logical Design

The following table outlines represent the logical design of our Key Space Website. All tables are normalized to Third Normal Form(3NF) To ensure data Integrity, eliminate redundancy, and support efficient web application performance.

## 1.BRANCH Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **BranchID** | **Integer** | **Primary key** | **Unique identifier for each branch office. Auto-incremented system-generated value.** |
| **BranchName** | **VARCHAR (100)** | | **The official trading name of the branch office (e.g., 'Key Space Johannesburg North').** |
| **City,** | **VARCHAR (100)** | | **The city in which the branch is physically located (e.g., Pretoria, Cape Town).** |
| **Province** | **VARCHAR (100)** | | **The province where the branch operates, used for regional filtering and reporting.** |
| **ContactNumber** | **VARCHAR (15)** | | **The branch's primary telephone or cell number for client and agent communications.** |

## 2.AGENT Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **AgentID** | Integer | Primary Key | Unique identifier for each real estate agent registered in the system. |
| ***BranchID*** | Integer | Foreign Key | References the BRANCH table. Identifies which branch this agent is employed by. |
| FirstName | VARCHAR(100) | | The agent's given (first) name, used in profiles, listings, and client communications. |
| LastName | VARCHAR (100) | | The agent's family (last) name, used together with FirstName for full identification. |
| ContactNumber | VARCHAR (15) | | Agent's direct contact number, provided to clients and used for appointment scheduling. |
| Email | VARCHAR (100) | | Agent's professional email address for correspondence with clients and the system. |
| Region | VARCHAR (100) | | The geographic region the agent operates in or specialises in (e.g., Sandton, Centurion). |
| ActiveListingCount | Integer | | 8.Derived attribute: the count of properties currently linked to this agent with status = 'available'. Calculated |

| | | | |
|---|---|---|---|
| | | | **automatically from the PROPERTY table.** |

## 3.LOCATION Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **LocationID** | **Integer** | **Primary key** | **1.Unique identifier for each location record. Referenced by properties to normalise address data.** |
| **PostalCode** | **Integer** | | **2.The postal code of the area, used for geographic filtering and postal delivery purposes.** |
| **City** | **VARCHAR (100)** | | **3.The city name associated with the location (e.g., Johannesburg, Durban).** |
| **Province** | **VARCHAR (100)** | | **4 .The province of the location, supporting region-based searches and reporting.** |

## 4.PROPERTY Table

| Field Name | Data type | Key | Description |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **PropertyID** | **Integer** | **Primary key** | **Unique identifier for each property listing in the system.** |
| ***AgentID*** | **Integer** | **Foreign Key** | **.References the AGENT table. Identifies the agent responsible for managing this listing.** |
| ***PropertyTypeID*** | **Integer** | **Foreign Key** | **References the PROPERTY_TYPE table. Classifies the property (e.g., residential, commercial).** |
| ***LocationID*** | **Integer** | **Foreign Key** | **.References the LOCATION table. Stores the normalised location details for the property.** |
| **Street** | **VARCHAR (100)** | | **The street address of the property. Together with Location fields, forms the composite address.** |
| **Title** | **VARCHAR (100)** | | **A short, descriptive headline for the listing (e.g., 'Spacious 3-Bed Home in Sandton').** |
| **Price** | **Decimal(12,2)** | | **.The listed sale or rental price of the property, stored with two decimal places for precision.** |
| **Size_sqm** | **Integer** | | **The total size of the property in square metres, used for filtering and comparison purposes.** |
| **Bedrooms** | **VARCHAR (50)** | | **The number of bedrooms in the property, stored** |

| | | | |
|---|---|---|---|
| | | | as text to support values such as 'Studio'. |
| Bathrooms | VARCHAR (50) | | The number of bathrooms in the property, stored as text to allow flexibility (e.g., '2.5'). |
| Status | VARCHAR (50) | | Current availability status of the property: 'available', 'sold', or 'rented'. |
| ListingDate | Date | | The date on which the property was listed on the platform, used for sorting and reporting. |

## 5.PROPERTY_TYPE Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| PropertyTypeID | Integer | Primary key | Unique identifier for each property classification type in the system. |
| TypeName | VARCHAR (100) | | The name of the property category (e.g., 'Residential', 'Commercial', 'Industrial', 'Land'). |
| Description | VARCHAR (100) | | A brief explanation of what the property type covers, providing context for agents and clients. |

## 6.CLIENT Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| ClientID | Integer | Primary key | Unique identifier for each registered client on the platform. |

| | | | |
|---|---|---|---|
| **FirstName** | **VARCHAR(100)** | | **The client's given name, used in all client-facing communications and records.** |
| **LastName** | **VARCHAR(100)** | | **The client's family name, used together with FirstName for full identification.** |
| **Email** | **VARCHAR(100)** | | **Client's email address, used for account login, notifications, and correspondence.** |
| **ContactNumber** | **VARCHAR(15)** | | **Client's phone or cell number, used for agent contact and appointment scheduling.** |
| **Client Type** | **VARCHAR(100)** | | **Categorises the client's role: 'buyer', 'seller', 'tenant', or 'landlord'.** |
| **BudgetMin** | **Decimal(12,2)** | | **The minimum amount the client is willing to spend or accept, used to filter matching properties.** |
| **BudgetMax** | **Decimal(12,2)** | | **The maximum amount the client is willing to spend, used as the upper limit for property matching.** |
| **PreferredLocation** | **VARCHAR(100)** | | **The geographic area or neighbourhood the client prefers when searching for properties.** |
| **PreferredPropertyType** | **VARCHAR(100)** | | **.The type of property the client is most interested in (e.g.,** |

| | | | |
|---|---|---|---|
| | | | **'Residential', 'Commercial').** |
| **RegistrationDate** | **Date** | | **The date the client created their account on the platform, used for auditing and CRM reporting.** |

## 7.TRANSACTION Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **TransactionID** | **Integer** | **Primary key** | **Unique identifier for each completed or in-progress transaction in the system.** |
| ***PropertyID*** | **Integer** | **Foreign Key** | **References the PROPERTY table. Identifies the property involved in this transaction.** |
| ***ClientID*** | **Integer** | **Foreign Key** | **References the CLIENT table. Identifies the client who is party to this transaction.** |
| ***AgentID*** | **Integer** | **Foreign Key** | **References the AGENT table. Identifies the agent who facilitated and closed the transaction.** |
| **TransactionType** | **VARCHAR(100)** | | **Indicates the nature of the transaction: 'sale' or 'rental'.** |
| **TransactionDate** | **VARCHAR(100)** | | **The date on which the transaction was formally completed or recorded in the system.** |

| | | | |
|---|---|---|---|
| **FinalPrice** | **Decimal(12,2)** | | **The agreed final amount of the transaction, which may differ from the original listed price.** |
| **Status** | **VARCHAR(50)** | | **Current state of the transaction: e.g., 'pending', 'completed', or 'cancelled'.** |

## 8. PROPERTY_FEATURE

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **FeatureID** | **Integer** | **Primary key** | **Unique identifier for each feature entry. Enables each feature to be stored and retrieved individually.** |
| ***PropertyID*** | **Integer** | **Foreign Key** | **References the PROPERTY table. Links the feature to its associated property listing.** |
| **FeatureName** | **VARCHAR(100)** | | **Describes the specific feature available at the property (e.g., 'Swimming Pool', 'Garage', 'Garden', 'Security System').** |

## 9. CLIENT_SAVED_PROPERTY

| Field Name | Data type | Key | Description |
|---|---|---|---|
| **ClientID** | **Integer** | | **References the CLIENT table. Combined with PropertyID to form the composite primary key. Identifies the client** |

| | | | |
|---|---|---|---|
| | | | who saved the property. |
| PropertyID | Integer | Foreign Key | References the PROPERTY table. Combined with ClientID forms the composite primary key. Identifies the saved property. |
| SavedDate | Date | | The date on which the client bookmarked or saved the property listing for future reference. |

## 10.INQUIRY Table

| Field Name | Data type | Key | Description |
|---|---|---|---|
| InquiryID | Integer | | Unique identifier for each inquiry submitted on the platform. |
| *ClientID* | Integer | Foreign Key | References the CLIENT table. Identifies the client who submitted the inquiry. |
| *PropertyID* | Integer | Foreign Key | References the PROPERTY table. Identifies the property the inquiry is about. |
| *AgentID* | Integer | Foreign Key | .References the AGENT table. Identifies the agent assigned to handle this inquiry. |
| TransactionType | VARCHAR(100) | | The type of inquiry submitted, such as 'Viewing Request', 'Price Enquiry', or 'General Information'. |
| TransactionDate | VARCHAR(100) | | The date on which the client submitted the inquiry. |
| FinalPrice | Decimal(12,2) | | Current status of the inquiry: e.g., 'pending', 'responded', or 'closed'. |

| | | | |
|---|---|---|---|
| **Status** | **VARCHAR(100)** | | **Any additional comments or context provided by the client at the time of submission** |